

Questionnaire English Translation

Study: Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision

This document is a faithful English translation of the original German questionnaire administered via Microsoft Forms. Item numbering follows the original. Question type and any selection limits are reproduced as displayed. Conditional display logic (branching) is not represented in the Microsoft Forms export and is therefore not captured here; see the notes on items 15, 16, 23 and 24. Spelling errors present in the German original have been silently corrected in translation. The English wording of items reported in the manuscript follows the wording used there.

Part 1: Consent to data protection and participation in the study

Thank you for taking part in the study 'Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision'.

The study consists of three sections: (1) consent to data protection and participation in the study; (2) socio-demographic information; (3) information on previous digitalisation and AI experiences and related training.

Answering the questions takes approximately 8 to 10 minutes in total.

If you have further questions or uncertainties, you may contact the study leader [name and institutional email redacted] at any time.

Q1. I have read the data protection declaration and agree to it.

Single choice. Required. A link to the data protection declaration was provided in the original [link redacted].

Consent statement shown to respondents: I have read the data protection declaration for participation in the study and consent to the associated data processing. I am aware that consent is voluntary and may be refused without disadvantage or withdrawn at any time without giving reasons. I understand that, in the event of withdrawal, the lawfulness of the processing carried out on the basis of consent up to that point is not affected, and that to withdraw I can simply contact the person named in the participant information and the data protection declaration.

Q2. I have read the participant information and consent to take part in the study.

Single choice. Required. A link to the participant information was provided in the original [link redacted].

Consent statement shown to respondents: I have read the participant information and the data protection declaration for participation in the study and consent to participation and the associated data processing. The voluntariness and withdrawal conditions stated for item 1 apply equally.

Part 2: Socio-demographic questions

Q3. How old are you?

Single choice. Required.

- 20 to 29 years
- 30 to 39 years
- 40 to 49 years
- 50 to 59 years
- 60 to 69 years
- 70 years or older

- No answer

Q4. In which federal state do you work?

Single choice. Required.

- Baden-Wuerttemberg
- Bavaria
- Brandenburg
- Berlin
- Bremen
- Hamburg
- Hesse
- Mecklenburg-Western Pomerania
- Lower Saxony
- North Rhine-Westphalia
- Rhineland-Palatinate
- Saarland
- Saxony
- Saxony-Anhalt
- Schleswig-Holstein
- Thuringia
- No answer
- Other

Q5. Which gender do you identify with?

Single choice. Required.

- Male
- Female
- Diverse
- No answer

Q6. In which setting do you mainly work?

Single choice. Required.

- Hospital (inpatient)
- Hospital (outpatient)
- Practice (single practice)
- Practice (group practice / joint medical practice)
- Medical care centre (MVZ)
- Nursing home / outpatient care
- Emergency medical service
- Public health service
- Administration / management
- Self-governance / politics / professional association
- No answer

Q7. Which professional group do you belong to?

Single choice. Required.

- Medical assistant (MFA)
- Professional nursing

- Physician
- Other healthcare professionals
- Other

Q8. In which specialty do you work?

Single choice. Required.

- General medicine
- General surgery
- Anaesthesiology
- Occupational medicine
- Ophthalmology
- Dermatology
- Gastroenterology
- Gynaecology
- Geriatrics
- Otorhinolaryngology (ENT)
- Haematology and oncology
- Human genetics
- Hygiene and environmental medicine
- Internal medicine
- Intensive care medicine
- Cardiology
- Paediatric and adolescent medicine
- Child and adolescent psychiatry and psychotherapy
- Laboratory medicine
- Oral and maxillofacial surgery
- Neurosurgery
- Neurology
- Nuclear medicine
- Public health
- Orthopaedics and trauma surgery (incl. plastic and reconstructive surgery)
- Pathology
- Pharmacology
- Psychiatry
- Psychosomatic medicine
- Radiology
- Forensic medicine
- Radiation medicine
- Thoracic, cardiac and vascular surgery
- Transfusion medicine
- Urology
- No answer
- Other

Q9. How much professional experience do you have? (Counted from the third state examination)

Single choice. Required. The reference point indicates that this item is addressed to physicians.

- 0 to 4 years

- 5 to 9 years
- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Q10. What is your job title / position? (Nursing and medical assistants)

Single choice. Required. Addressed to nursing staff and medical assistants.

- Geriatric nurse
- Trainee medical assistant (MFA)
- Nursing trainee
- Medical assistant (MFA) / doctor's assistant
- Lead medical assistant
- Medical assistant with additional qualification (e.g. VERAH, NaePa, practice manager, digital manager)
- Paramedic or emergency medical service staff
- Nursing assistant or nursing aide
- Registered nurse (health and nursing care professional)
- Practice instructor in nursing
- Specialist nurse with further training (e.g. intensive care, anaesthesia, operating theatre, paediatric intensive care)
- Physician assistant
- Ward manager or team leader
- Nursing service manager
- Other

Q11. How much professional experience do you have? (Counted from completion of training / studies)

Single choice. Required. Addressed to nursing staff and medical assistants.

- 0 to 4 years
- 5 to 9 years
- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Part 3: Questions on digitalisation and AI experiences and related training

Definition shown to respondents: There is no uniform definition of artificial intelligence. For this questionnaire we therefore rely on the definition of the Deutsches Ärzteblatt, namely that AI is a subfield of computer science and the ability of systems to perform independently cognitive functions such as learning, thinking, planning and problem-solving that normally require human intelligence, while (still) differing from human intelligence in central characteristics such as consciousness. Source: Deutsches Ärzteblatt, AI special issue 2025.

Q12. How would you rate your digital competence in everyday life (non-professional context)?

Rating scale 1 to 5, where 1 = very low and 5 = very high.

Q13. How frequently do you use the following digital tools in your working day?

Matrix question. One frequency selected per row. Frequency categories: daily, weekly, monthly.

Tool	Daily	Weekly	Monthly
Practice management systems (PVS) / hospital information system (HIS)			
Electronic patient record (ePA)			
Electronic medication list (eML / eMP)			
Electronic certificate of incapacity for work (eAU)			
Image viewing systems (e.g. PACS)			
Laboratory information systems (LIS)			
Differential diagnostics / clinical decision support (CDSS)			
Digital health applications (DiGA)			
Health apps for patient care			
Telephone assistance systems (e.g. for appointment scheduling)			
Video consultation			
Tele-visit (e.g. intensive care unit / stroke unit / nursing home)			
Tele- / home monitoring			
Speech recognition for automated treatment documentation			
Digital patient pre-registration (NIDA, IVENA, Pulsation) in the emergency department / emergency medical service			
Initial assessment / triage tools (e.g. from docyet, ada, SMeD)			
Drug therapy safety system (AMTS): alerts when prescribing medication			
Automated duty / shift scheduling			
Knowledge management (e.g. Amboss, Elsevier, Thieme)			

Q14. Does your employer provide you with AI-based applications?

Single choice. Required. This item is the basis for the governance index component 'provision of AI tools' reported in the manuscript.

- Yes
- No

- I don't know / cannot say

Q15. In what form are these AI tools provided?

Multiple choice. Required. Conditional on item 14; displayed where AI tools are provided.

- As standalone tools or programs (e.g. AI models or chatbots, speech recognition, etc.)
- Integrated into existing medical software (e.g. diagnostic systems, decision support)
- In trial / pilot phase (external software)
- In-house developed AI models
- No AI tools available
- Don't know
- Other

Q16. What are the main reasons why you currently do not use AI-based applications?

Multiple choice, select at most 3. Required. Conditional; addressed to respondents who do not use AI applications.

- Not available
- Security / data protection concerns
- Lack of knowledge / training
- No perceived usefulness / time saving
- Lack of integration options in existing systems
- Lack of capacity / time for implementation
- Other

Q17. How would you rate your digital competence in a professional context?

Rating scale 1 to 5, where 1 = very low and 5 = very high. Required.

Q18. How do you rate the support (e.g. training, instructions) when introducing new digital tools at your workplace?

Free text. Required.

Q19. Do you already use AI-based applications in your professional daily routine?

Single choice. Required. Provides the AI-use frequency categories reported in the manuscript.

- Yes, regularly (several times per week)
- Yes, occasionally (once per week)
- Yes, rarely / experimentally (once per month)
- No, but planned / interested
- No, and not currently planned
- I don't know / am unsure
- No answer

Q20. For which of the following areas do you use AI applications? (Multiple answers possible)

Multiple choice. Required. Provides the application-area figures reported in the manuscript.

- Diagnostics and findings (e.g. image analysis, pattern recognition, ECG evaluation, etc.)
- Treatment planning and therapy decisions (e.g. medication management, personalised therapy recommendations)
- Administrative tasks (e.g. documentation such as writing physician letters, billing, etc.)
- Research (e.g. data analysis, literature search, simulations)

- Patient communication and service (e.g. chatbots, appointment scheduling, automated history-taking tools)
- No answer
- Other

Q21. How do you engage with the topic of AI in your private life? (Multiple answers possible)

Multiple choice. Required.

- I actively inform myself (e.g. specialist articles, courses, podcasts)
- I occasionally read news / articles about it
- I use AI applications privately (e.g. ChatGPT, AI image generators)
- Via social media or networks (e.g. LinkedIn)
- Hardly at all / only passively
- No answer
- Other

Q22. Have you already received training specifically on the topic of AI in a professional context?

Single choice. Required. This item is the basis for the governance index component 'AI-specific training'.

- Yes
- No
- No answer

Q23. What type of AI training was it? (Multiple answers possible)

Multiple choice. Required. Conditional on item 22; displayed where training was received.

- Short presentation / information session (under 2 hours)
- Workshop (a few hours up to 1 day)
- Multi-day course / certificate programme
- Online module / e-learning
- Integrated into other continuing education
- No answer
- Other

Q24. How do you rate the usefulness of this training for your work?

Rating scale 1 to 5, where 1 = not useful at all and 5 = very useful. Required. Conditional on item 22.

Q25. Which formats for AI training would you prefer? (Multiple answers possible)

Multiple choice. Required.

- Online courses (flexible, self-study format)
- Live online webinars (interactive)
- In-person events / on-site workshop
- Blended learning (combination of online and in-person)
- Short information nuggets (e.g. 15 to 30 minute videos)
- No preference
- Other

Q26. What scope / duration for AI training would you consider ideal? (Multiple answers possible)

Multiple choice. Required.

- Short formats (e.g. 1 to 2 hours)
- Half-day events
- Full-day events
- Multi-day courses
- Longer-term continuing education (e.g. spread over several weeks / months)
- No preference
- Other

Q27. At what level should the AI training ideally be pitched for you? (Multiple answers possible)

Multiple choice. Required.

- Introductory / foundational (focus on understanding basic concepts)
- Application-oriented / practical (focus on using specific tools and methods in daily work)
- Advanced / specialised (focus on detailed knowledge of specific AI methods or complex use cases)
- Strategic / management-oriented (focus on implementation)
- No answer
- Other

Q28. Which of the following topic areas for AI training would be particularly relevant for you?

Multiple choice, select at most 3. Required.

- AI in diagnostic imaging
- AI-supported diagnostic systems
- AI in therapy planning and personalised medicine
- Handling large language models (e.g. ChatGPT) in the medical context
- Data management and data protection in AI applications
- Ethical and legal frameworks for AI in medicine
- Critical appraisal of AI applications and studies
- Practical implementation of AI tools in daily work
- AI in medical research
- AI and patient interaction and communication
- No answer
- Other

Q29. Which organisation or institution should provide this offering? (Multiple answers possible)

Multiple choice. Required.

- Medical chambers (Ärzttekammern)
- Employers (hospitals / medical practices)
- Professional associations (e.g. Hartmannbund e.V., Marburger Bund)
- Specialist societies (e.g. DGIM, DEGAM, DGOU)
- Industry providers of AI software
- Associations of Statutory Health Insurance Physicians (Kassenaerztliche Vereinigungen)
- Independent private continuing education institutes
- Universities / higher education institutions
- No preference
- Other

Q30. What reasons keep you from pursuing further training on AI in medicine?

Free text. Required.

Q31. Do you have any further comments or requests regarding AI training for your professional group?

Free text. Required.

Q32. Are you aware of cases of Shadow AI in your professional environment?

Single choice. Required. This is the core Shadow AI outcome item. The single-choice format does not allow respondents to report both own use and observation of others; the manuscript therefore treats the exposure rate as a lower bound.

Definition shown to respondents: Shadow AI refers to the use of AI-based applications or tools by physicians without official authorisation or without oversight by the responsible IT or data protection officers. Example: using ChatGPT on a private device to make a diagnosis.

- Yes, I have used AI myself without formal authorisation before.
- Yes, I have seen others use AI without formal authorisation before.
- No, I am not aware of any such case.
- No answer

Q33. Have you received a guideline or similar from your employer on the deployment and use of artificial intelligence, in particular large language models (ChatGPT, Microsoft Co-Pilot, Claude, etc.)?

Single choice. This item is the basis for the governance index component 'written guidelines'.

- Yes
- No
- I don't know
- Other